TMCFN: Text-Supervised Multidimensional Contrastive Fusion Network for Hyperspectral and LiDAR Classification

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***Abstract*— The joint classification of hyperspectral images (HSIs) and LiDAR data plays a crucial role in Earth observation missions. Most advanced methods are based on discrete label supervision. However, since discrete labels only convey limited information that a sample belongs to a single definite class and lack of prior information, it is difficult to supervise the model to capture rich inherent semantic information in complex data distributions, hindering the classification performance. To this end, we propose a text-supervised multidimensional contrastive fusion network (TMCFN), which leverages class text information to guide the learning of visual representations while establishing a semantic association of text and visual features for classification using multidimensionally incorporated contrastive learning (CL) paradigms. Specifically, TMCFN is composed of text information encoding (TIE), visual features representation (VFR), and text–visual features alignment and classification (TVFAC). TIE is employed to extract semantic information from class text extended from class names, intrinsic attributes and interclass relationships. VFR mainly comprises a new fusion-based contrastive feature learning module (FCFLM) to extract discriminative visual features and a text-guided attention feature fusion module (TAF^{2}M) to fuse visual features under the guidance of text information. TVFAC optimizes the learning of visual features under the supervision of text information while using a CL paradigm to align text and visual features for establishing the semantic association, and achieves the classification by directly computing the similarity between the visual features and each text feature without an additional classifier. Experiments with three standard datasets verify the effectiveness of TMCFN.**

***Index Terms*— Contrastive learning (CL), hyperspectral images (HSIs), image classification, LiDAR, multimodal remote sensing.**

Manuscript received 16 December 2023; revised 13 February 2024; accepted 4 March 2024. Date of publication 7 March 2024; date of current version 18 March 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62101414 and Grant 62201423, in part by the Young Talent Fund of Xi'an Association for Science and Technology under Grant 095920221320 and Grant 959202313052, in part by China Postdoctoral Science Special Foundation under Grant 2022T150508 and Grant 2023T160502, in part by the Young Talent Fund of Association for Science and Technology in Shaanxi under Grant 20230117, in part by the Youth Innovation Team of Shaanxi Universities, in part by China Postdoctoral Science Foundation under Grant 2021M702546 and Grant 2021M702548, in part by the Foundation of Anhui Province Key Laboratory of Physical Geographic Environment under Grant 2023PGE02, and in part by the Open Research Fund of Shaanxi Key Laboratory of Optical Remote Sensing and Intelligent Information Processing under Grant KF20230302. *(Corresponding authors: Jiahui Qu; Wenqian Dong.)*

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Digital Object Identifier 10.1109/TGRS.2024.3374372

I. Introduction

With the advancement of satellite remote sensing technology, remote sensing images have become more accessible. Hyperspectral images (HSIs) [1], [2], [3], [4] are captured by imaging spectrometers, which is capable of collecting extensive narrow-band spectral information across the visible to infrared spectra that is beneficial for distinguishing ground objects. However, HSIs face challenges in identifying ground objects with same spectral characteristics but varying heights. In contrast, LiDAR data [5], [6] provide detailed spatial and elevation information within an area and are less susceptible to weather conditions (e.g., cloudy or rainy). Nonetheless, it is difficult to differentiate ground objects with different materials and the same height when relying solely on LiDAR data. Joint classification of HSIs and LiDAR data can make full use of their complementary information to improve the accuracy of classification, and is widely used in environmental monitoring [7], [8], [9] urban planning [10], agriculture [11], [12], and so on.

In the recent past, many machine learning techniques have been applied to joint classification of HSIs and LiDAR data, including support vector machine (SVM) [13], random forests [14], rotation forests (RoFs) [15], and so on. Pedergnana et al. [16] fuse and classify features extracted by calculating extended attribute profiles (EAPs) on optical images and LiDAR data, achieving the integration of spectral, spatial, and elevation information. Orthogonal total variation component analysis (OTVCA) [17] is introduced to integrate spatial–elevation features extracted from HSIs and LiDAR data using extinction profiles (EPs) with the spectral features derived from only HSIs, generating the final feature map. Rasti et al. [14] propose a sparse and low-rank technique to fuse spatial and elevation information extracted from HSIs and LiDAR data using EPs, leading to encouraging fusion results. Zhang et al. [18] utilize diverse classifiers, such as SVM, RF, and KNN, to derive joint classification results and accomplish a more robust fusion classification at the decision level through a majority-weighted voting. However, traditional methods of applying these techniques rely on the quality of the handcrafted features, which limits their application scenarios.

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Recently, deep learning-based methods have gained attention owing to their exceptional abilities in automated features extraction. Convolutional neural networks (CNNs) stand out as a robust methodology in the realm of remote sensing image processing [19], [20], [21], [22], owing to their distinctive receptive field and the parameter sharing mechanism inherent in the convolution kernels. Zhan et al. [23] introduce an enhanced spectral–spatial residual attention network, which extracts more discriminative spectral and spatial features and improves model classification performance. Wang et al. [24] utilize the feature cross-learning stream fusion to make full use of the information that supplements each other of HSIs and LiDAR data, effectively improving the classification accuracy. The transformer structure employs the self-attention mechanism to extract features [25], effectively addressing the distance-related challenges in sequence data processing within the domain of natural language processing. Simultaneously, numerous studies have incorporated the advantages of the transformer into the realm of remote sensing image processing [26], [27], [28]. Qiu et al. [26] propose a cross-channel dynamic spatial–spectral fusion transformer, which extracts multiscale features, multichannel features, and cross-channel global features, improving classification accuracy. Zhao et al. [27] propose a dual-branch method comprising hierarchical CNNs and transformer networks to extract spatial–spectral features from HSIs and elevation-spatial features from LiDAR data. Subsequently, a cross-token attention (CTA) fusion encoder is employed to merge these features, achieving remarkable classification results. Roy et al. [28] present a multimodal fusion transformer (MFT) network, which employs a multihead cross-patch attention (mCrossPA) for extracting complementary information from HSIs and LiDAR data, achieving better generalization. Additionally, numerous approaches employ graph neural networks (GNNs) to capture the topological relationships between pixels [29], [30], [31], [32], providing a more comprehensive and accurate feature representation for classification tasks. Li et al. [31] introduce GNNs to obtain high-confidence pseudolabels, facilitating the semi-supervised training of the model. Wang and Wang [32] use CNNs and GNNs to extract spatial–spectral features and structural features from short-range and middle-to-long-range regions of HSIs and LiDAR data, respectively, and fuse these complementary features, markedly improving classification precision.

The aforementioned methods are all based on discrete label supervision, where the model outputs a discrete class value and a loss function is utilized to establish the connection between the predicted value and the discrete label value for classification. However, discrete labels only convey the information that samples belong to a single definite class, and lack of prior information. These limit the model to capture the rich inherent semantic information in complex data distributions, hindering classification performance. In practice, the annotation information of data usually includes discrete label values along with their corresponding class names. In addition to the widely mined label values, the text information in class names has also been proven to be effective as a supervisory information to guide the training. Compared to discrete label supervision methods, text supervision methods offer the following advantages.

1) In contrast to discrete label supervision that provides fixed discrete class labels, text supervision can offer a more diverse information. Additionally, as a new modality, text features can effectively interact with visual features to facilitate the learning of visual representations for classification tasks.
2) By leveraging text supervision, we can effectively map the data into a semantic space, enabling the model to better capture semantic similarities and differences among the data, improving classification performance. However, how to effectively utilize text information to supervise model training in the classification tasks is still an open issue.

In this article, we propose a text-supervised multidimensional contrastive fusion network (TMCFN), which leverages class text information to guide the learning of visual representations while establishing a semantic association of text and visual features for classification. The proposed TMCFN (Fig. 1) consists of three key components: text information encoding (TIE), visual features representation (VFR), and text–visual features alignment and classification (TVFAC). In TIE, we extract rich text semantic information from extended class text consisting of class names, intrinsic attributes, and inter-class relations. Moreover, VFR predominantly depends on a fusion-based contrastive feature learning module (FCFLM) and a text-guided attention feature fusion module (TAF^{2}M) for the acquisition of visual features. Last but not least, TVFAC optimizes VFR based on text supervision while using a CL paradigm to align text and visual features for establishing the semantic association, and achieves the classification by directly computing the similarity between the visual features and each text feature without an additional classifier.

In summary, our main contributions are threefold.

1) We propose a TMCFN, which utilizes text information as a supervisory signal to guide the learning of visual representations while establishing semantic connections between text and visual features for classification.
2) We design a FCFLM in VFR to extract discriminative spatial–spectral–elevation features from HSIs and LiDAR data for classification based on a fusion-contrast perspective, fully exploiting potential multimodal information in available data.
3) We enrich class text with prior knowledge including intrinsic attributes and interclass relationships, and propose a TAF^{2}M that employs rich text semantic information to guide the fusion of visual features during the late fusion stage of VFR, promoting the complementary integration of multimodal information and facilitating the model to emphasize crucial features relevant to the classification task.

The remaining sections of this article are structured as follows. Some related works are presented in Section II. The proposed method is elaborated upon in Section III. Comprehensive experimental results and analysis, including comparative experiments and ablation studies, are discussed

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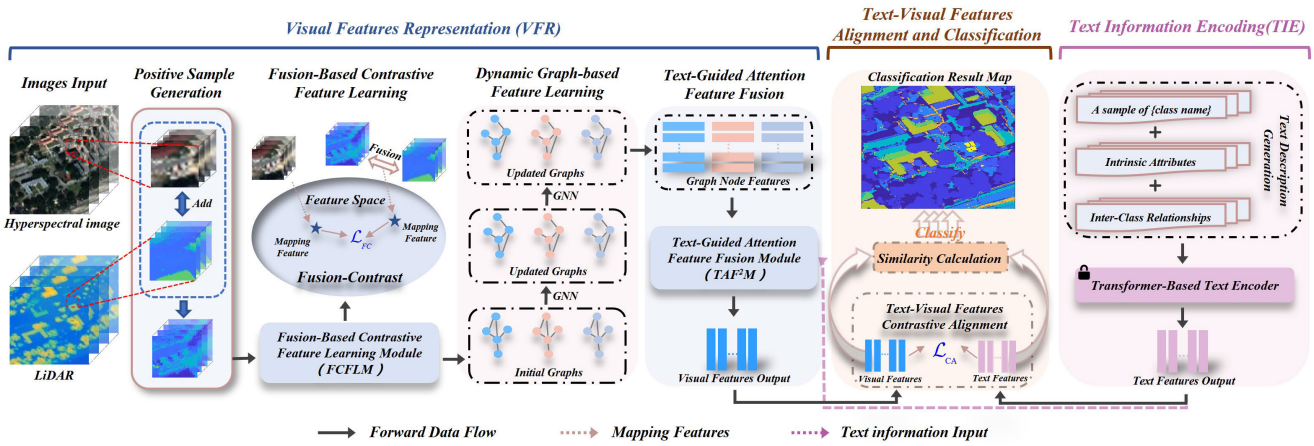


Fig. 1. Overview of the proposed TMCFN, which leverages class text information to guide the learning of visual representations while establishing a semantic association of text and visual features for classification. TMCFN consists of TIE, VFR, and TVFAC. TIE and VFR are responsible for extracting text features and multimodal features of vision, respectively. TVFAC establishes semantic connections between the text and visual features through CL and computes the similarity of visual features and each text feature to achieve classification without an additional classifier.

in Section IV. The conclusion is summarized with some concluding remarks in Section V.

II. Related Work

A. Representation Learning With Text Supervision

Contrastive language-image pretraining (CLIP) serves as a prominent exemplar of text-supervised representation learning model and has gained significant interest. It aligns text and images in a shared representation space through the CL, thereby establishing semantic correlations between texts and images. The learned representations by CLIP and its variants have been empirically proven to be effective in various downstream tasks, such as object detection [33], [34] and semantic segmentation [35], [36]. In this article, we follow the idea of CLIP to align text and visual features, and redesign the encoding process of visual features for efficiently encoding and fusing HSIs and LiDAR data.

B. Contrastive Learning (CL)

CL is a self-supervised learning technique that aims to extract meaningful representations from unlabeled data. Several well-known CL frameworks, such as SimCLR [37], MoCo [38], BYOL [39], and SimSiam [40], have been developed to learn an encoder with strong representation capabilities that can be transferred to downstream tasks with promising results. In contrast to SimCLR and MoCo, BYOL, and SimSiam do not require explicit negative pairs during training, and they train the encoder by predicting one view from another, where both views are derived from different augmentations of the same image. Note that the encoders obtained from the aforementioned frameworks are designed without a specific task. To achieve optimal performance on downstream tasks, it typically requires fine-tuning in the second stage, increasing the overall complexity of the algorithm. In this article, we design a BYOL-based module named FCFLM to extract visual features and embed it into the model for end-to-end training. Furthermore, we adopt the idea of CL to align text and visual features for establishing the connection between them.

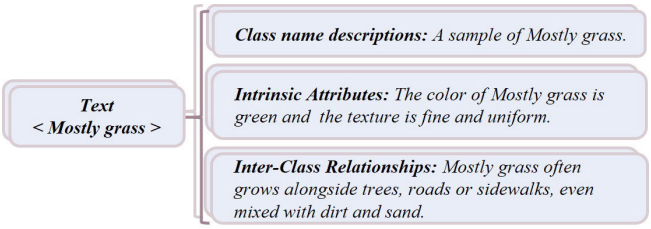


Fig. 2. Examples of class text descriptions. A class text description mainly includes the class name description, intrinsic attributes, and interclass relationships.

III. Method

Compared to discrete label values, class text can convey more comprehensive semantic information, which helps the model better capture the rich inherent semantic information in complex data distributions to improve classification performance. In this article, we propose a TMCFN, which leverages extended class text information to guide the learning of visual representations and attempts to utilize text features to interact with visual features for facilitating multimodal information fusion, while establishing semantic associations of text and visual features for classification using multidimensionally incorporated CL paradigms.

A. Text Information Encoding

In this section, we utilize prior information to extend the content of each category description, generating rich text descriptions. Subsequently, we tokenize and encode these text descriptions to obtain text features.

1) Text Description Generation: It is known that a class name usually consists of one or several words. It is difficult to extract contextual semantic features from such text. Therefore, we extend each class text, which includes class name descriptions, intrinsic attributes, and interclass relationships. Examples are depicted in Fig. 2. The class name descriptions

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follow a fixed template *a sample of {class name}*, providing a consistent representation, such as *a sample of Mostly grass*. Intrinsic attributes include color, texture characteristics, and other relevant characteristics that are specific to the class, such as *the color of mostly grass is green and the texture is fine and uniform*. Interclass relationships describe potential associations between samples of different categories, such as *mostly grass often grows alongside trees, roads, or sidewalks, even mixed with dirt and sand*.

2) Text Encoding: To enhance the overall effectiveness of the text encoding process, we start with utilizing lower cased byte pair encoding (BPE) [41] to obtain tokenized representations. Subsequently, a transformer [42] is employed to encode the tokenized representations and generate c classes text features $\mathbf{T}$. These text feature vectors, serving as text semantic information, are utilized to guide the fusion of visual features in the VFR.

B. Visual Features Representation

In this section, we mainly focus on the acquisition of visual features. To achieve this goal, we comprehensively consider the following three aspects: the extraction of latent fused information from multimodal images, the extraction of features from middle- and long-distance pixels, and the fusion of multimodal image features, which are achieved through carefully designed fusion-based contrastive feature learning, dynamic graph-based feature learning, and text-guided attention feature fusion, respectively.

Let $\mathbf{H} \in \mathbb{R}^{h\times w\times C_H}$ and $\mathbf{L} \in \mathbb{R}^{h\times w\times 1}$ represent the original HSIs and LiDAR data, respectively, where C_H represents the channel dimension of HSIs, h and w denote the height and width of both data, respectively. To reduce dimensionality of $\mathbf{H}$ and make the dimension of $\mathbf{L}$ consistent with $\mathbf{H}$, we perform principal component analysis (PCA) [43] on $\mathbf{H}$ and replicate the band on $\mathbf{L}$. With the goal of enhancing the capability to extract local features, we generate patches for all labeled pixels as well as partial unlabeled pixels, yielding HSIs patches and their corresponding LiDAR patches. To enable batch computation in training stage, the labeled training patches are evenly partitioned into M batches based on their respective classes, while the unlabeled patches are randomly and uniformly distributed among the previously established M batches. We randomly select a batch for training each time. $\mathbf{H}_v^1$ and $\mathbf{H}_v^2$ represent the HSIs and LiDAR data of the batch, respectively.

1) Fusion-Based Contrastive Feature Learning: CL is widely employed due to its ability to learn feature representations with high generalization without labeled information. As a representative of CL, BYOL learns representations by bringing augmented views from an image closer. However, BYOL was initially designed for a unimodal image and may not be applicable to multimodal remote sensing image classification tasks because it mainly extracts invariant representations of images across different modalities, but does not fully capture latent fused information between multimodal images that is beneficial for improving the classification accuracy. Therefore, according to BYOL, we design a FCFLM with a fusion-contrast perspective that learns latent fused representations between multimodal images by bringing each view closer to the fusion view.

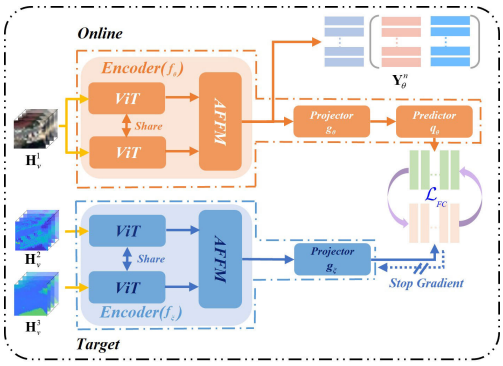


Fig. 3. Architecture of proposed FCFLM, which extracts latent spatial–spectral–elevation features. FCFLM inputs a group of positive samples (views) generated by a positive sample generation strategy, designs a new multimodal encoder ($f_{\theta/\xi}$) to obtain fusion views, and enhances multimodal representation learning by bringing each view closer to the fusion view.

The structure of FCFLM is shown in Fig. 3. As can be seen, the structure of FCFLM is similar to BYOL, including an online network (O-Net) and a target network (T-Net). O-Net consists of an encoder (f_θ), a projector (g_θ), and a predictor (q_θ), while T-Net also comprises the same encoder (f_ξ) and the same projector (g_ξ) except for the predictor.

There are two differences between FCFLM and BYOL. First, FCFLM and BYOL obtain positive samples in different ways. BYOL obtains positive samples through data augmentation. Different from BYOL, FCFLM believes that images with different modalities in the same area can be regarded as views (positive samples) with natural augmentation, where the feature information of different modalities can be fully mined. Moreover, given the significant modality difference between the HSIs and LiDAR data, it is difficult for the encoder to converge when only using $\mathbf{H}_v^1$ and $\mathbf{H}_v^2$ as positive sample pairs. To address this issue, we propose a positive sample generation strategy to obtain $\mathbf{H}_v^3$, which is generated by adding $\mathbf{H}_v^1$ and $\mathbf{H}_v^2$. $\mathbf{H}_v^3$ serves as an intermediary data and a positive sample for both $\mathbf{H}_v^1$ and $\mathbf{H}_v^2$, effectively bridging the gap between positive samples with different modalities. Second, unlike BYOL which utilizes a standard unimodal encoder, FCFLM employs a novel multimodal encoder ($f_{\theta/\xi}$) to achieve the fusion of views, and learns latent fused representations between multimodal images by bringing each view closer to the fusion view. The multimodal encoder consists of a shared ViT utilized to capture both global and local features within patches, along with a meticulously designed attention feature fusion module (AFFM) to fuse ViT-encoded features for subsequent CL. In AFFM, we denote the two visual features inputs to the module as $\mathbf{F}_1$ and $\mathbf{F}_2$, respectively. The process of obtaining fusion feature $\mathbf{Y}$ is summarized as follows:

$$\alpha = \text{Sigmoid}\left[\mathbf{F}_1\mathbf{W}_1 \otimes (\mathbf{F}_2\mathbf{W}_2)^T\right] \tag{1}$$

$$\mathbf{Y} = \{\mathbf{F}_1 \parallel [\alpha\mathbf{F}_1 + (1-\alpha)\mathbf{F}_2] \parallel \mathbf{F}_2\}\mathbf{W}_3 \tag{2}$$

where $\otimes$ denotes matrix multiplication, $\parallel$ refers to operation of concatenation along the channel dimension, $\mathbf{W}_1$, $\mathbf{W}_2$, and $\mathbf{W}_3$ denote the weight of linear layers, respectively.

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Based on fusion-contrast perspective and assuming the input to the FCFLM are $\{\mathbf{H}_v^1, \mathbf{H}_v^2, \mathbf{H}_v^3\}$, we can infer that O-Net produces feature representations $\mathbf{Y}_\theta \triangleq f_\theta(\mathbf{H}_v^1, \mathbf{H}_v^1)$, a projection $\mathbf{Z}_\theta \triangleq g_\theta(\mathbf{Y}_\theta)$ and a predictor $q_\theta(\mathbf{Z}_\theta)$, while T-Net generates $\mathbf{Y}_\xi \triangleq f_\xi(\mathbf{H}_v^2, \mathbf{H}_v^3)$ and a projection $\mathbf{Z}_\xi \triangleq g_\xi(\mathbf{Y}_\xi)$. Subsequently, we utilize a mean-squared error loss function to maximize the consistency between features extracted from the views of the same area, facilitating the encoder to acquire more generalized feature representations. The loss function is expressed as follows:

$$\mathcal{L}_{\theta,\xi} \triangleq \left\|\bar{q}_\theta(\mathbf{Z}_\theta) - \bar{\mathbf{Z}}_\xi\right\|_2^2 = 2 - 2 \cdot \frac{\left\langle q_\theta(\mathbf{Z}_\theta), \mathbf{Z}_\xi \right\rangle}{\|q_\theta(\mathbf{Z}_\theta)\|_2 \cdot \left\|\mathbf{Z}_\xi\right\|_2} \tag{3}$$

where $\bar{q}_\theta(\mathbf{Z}_\theta) \triangleq q_\theta(\mathbf{Z}_\theta)/\|q_\theta(\mathbf{Z}_\theta)\|_2$ and $\bar{\mathbf{Z}}_\xi \triangleq \mathbf{Z}_\xi/\|\mathbf{Z}_\xi\|_2$ represent l_2-normalized $q_\theta(\mathbf{Z}_\theta)$ and $\mathbf{Z}_\xi$, respectively. We sequentially input one of $\{\mathbf{H}_v^1, \mathbf{H}_v^2, \mathbf{H}_v^3\}$ into O-Net and the remaining two samples into T-Net, concluding the training for this group of data when each sample has been processed once by O-Net. Accordingly, we can derive fusion-contrast loss as follows:

$$\mathcal{L}_{FC} = \sum_{i=1,2,3} \mathcal{L}_{\theta,\xi}^i. \tag{4}$$

With the shortcoming of the two-stage training method, we seamlessly integrate FCFLM into the model during training to simplify the training process, and we employ a unified loss function to optimize only θ, while ξ is a slowly moving exponential average of $\xi \leftarrow \tau\xi + (1-\tau)\theta$, where τ is a decay rate.

2) Dynamic Graph-Based Feature Learning: Throughout the encoding process of FCFLM, the ViT encoder captures both local and global features of the patches, essentially representing short distance features between pixels throughout the whole image. Considering the limitations of the distance within the feature context, we introduce the GNNs to strengthen the modeling of middle- and long-distance features between pixels throughout the whole image, capturing local and global topological relationships, especially in complex interactions between pixels. First, we apply a label mask to features $\mathbf{Y}_\theta^n$ with $n \in [1, 3]$, which are obtained from FCFLM, deriving labeled features for each batch. An initial adjacency matrix $\mathbf{A}$ is obtained by the following formulas:

$$\mathbf{X} = \mathbf{Y}_\theta^1 \parallel \mathbf{Y}_\theta^2 \parallel \mathbf{Y}_\theta^3 \tag{5}$$

$$\mathbf{A}^{i,j} = \|\mathbf{x}_i - \mathbf{x}_j\|_2 \tag{6}$$

where (6) calculates the Euclidean distance of two features vector $\mathbf{x}_i$ and $\mathbf{x}_j$ of $\mathbf{X}$. By this step, we can use $\mathbf{Y}_\theta^n$ and a shared $\mathbf{A}$ to get three graphs.

To further enhance the ability of GNNs to capture complex relationships and information within data, we design a dynamic graph structure generation module (DGSGM) modified from DGM [44] to update graph structures for dynamically extracting structural features. In DGSGM, the node embeddings of each graph are concatenated along the channel dimension to generate a new graph embeddings $\mathbf{X}'$. GNNs denoted as f_ϕ map $\mathbf{X}'$ to a lower dimensional space, producing auxiliary features $\tilde{\mathbf{X}}$ for computing edge probabilities $\mathbf{P}$, which can be formulated as follows:

$$\mathbf{P}_{i,j}\left(\tilde{\mathbf{X}}; \phi, t\right) = e^{-t \cdot D(\tilde{\mathbf{x}}_i, \tilde{\mathbf{x}}_j)} = e^{-t \cdot D(f_\phi(\mathbf{x}_i), f_\phi(\mathbf{x}_j))} \tag{7}$$

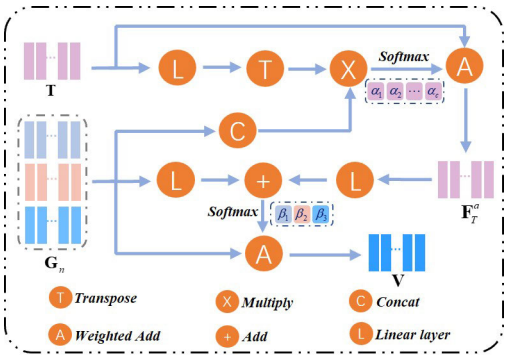


Fig. 4. Architecture of our proposed TAF^{2}M, which utilizes text information to adaptively adjust the contributions among visual features from images of different modalities, obtaining fused features of vision.

where $\mathbf{x}_i$ and $\mathbf{x}_j$ denote features vectors of $\mathbf{X}'$, t represents a trainable parameter, and $D(\cdot,\cdot)$ denotes the Euclidean distance between two nodes in the graph embedding space. Subsequently, a sampling method based on the Gumbel-Top-k trick [45] is applied to $\mathbf{P}$ to sample edges and obtain the edge set ε, which is essentially the new adjacency matrix $\mathbf{A}'$. Ultimately, the output features $\mathbf{G}_n$ of dynamic graph-based feature learning are obtained as follows:

$$\mathbf{G}_n = \mathrm{GNN}\big(\mathrm{GNN}\big(\mathbf{Y}_\theta^n, \mathbf{A}\big), \mathbf{A}'\big), n \in [1, 3]. \tag{8}$$

3) Text-Guided Attention Feature Fusion: To promote the complementary integration of visual features $\mathbf{G}_n$, we propose a TAF^{2}M, which leverages rich text semantic information to guide the fusion of visual features with different modalities. It is worth noting that TAF^{2}M employs weighted text features to guide the learning of visual features instead of original text features $\mathbf{T}$. The former are acquired by adaptively focusing on the key information of class text corresponding to the visual features, offering enhanced guidance for the subsequent fusion of visual features. In addition, TAF^{2}M utilizes a text-guided attention operation to dynamically calculates the weights among visual features from multimodal images under the guidance of text information, leading to an improved fusion of multimodal image features.

The architecture of TAF^{2}M is shown in Fig. 4. We initially input visual features $\mathbf{G}_n$ and text features $\mathbf{T}$. Given the category-agnostic nature of input visual features, it is not possible to directly specify a particular category of text semantic information to guide the fusion of visual features. Accordingly, we apply the following formula to obtain weighted text features $\mathbf{F}_T^a$, which adaptively emphasizes pivotal information of the class text corresponding to the visual features:

$$\mathbf{F}_T^a = \left\{\mathrm{Softmax}\left[(\mathbf{G}_1 \parallel \mathbf{G}_2 \parallel \mathbf{G}_3) \otimes (\mathbf{T}\mathbf{W}_4)^T\right]\right\} \otimes \mathbf{T} \tag{9}$$

where $\mathbf{W}_4$ denotes the weight of linear layers. Subsequently, a text-guided attention operation is employed to flexibly prioritize the aspects of visual features that are more crucial for the current classification task, contributing to more comprehensive visual features $\mathbf{V}$ with low redundancy as follows:

$$\beta_i = \mathrm{Attenton}\big(\mathbf{G}_i, \mathbf{F}_T^a\big)$$

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$$= \text{Softmax}\left\{\mathbf{G}_i\mathbf{W}_5 + \mathbf{F}_T^a\mathbf{W}_6\right\}, i \in [1,3] \tag{10}$$

$$\mathbf{V} = \sum_{i=1,2,3} \beta_i \cdot \mathbf{G}_i \tag{11}$$

where $\mathbf{W}_5$ and $\mathbf{W}_6$ denote the weight of linear layers, respectively.

C. Text–Visual Feature Alignment and Classification

In this section, we align text features $\mathbf{T}$ and visual features $\mathbf{V}$ for establishing semantic associations by a CL paradigm, and classify the visual features. In the alignment process, we select l visual features from each class for training. Unlike the general CLIP, where a single text is typically associated with an image, our scenario involves multiple pixels corresponding to a class text description. Consequently, we randomly choose a visual features per class from the training set, obtaining a total of $l * c$ text–visual feature pairs for per batch, enabling our model to benefit from a richer training set and learning robust associations between text and visual representations. We maximize the cosine similarity of c true text–visual features pairs while minimizing the cosine similarity of $c^2 - c$ incorrect pairs, establishing semantic associations of text and visual features for classification.

During the classification process, all visual features will calculate cosine similarity with text features of each category, scaled by temperature parameter τ, then normalized to probability distribution by softmax function. The predicted vector y can be obtained with the following formula:

$$y = \text{Argmax}\{\text{Softmax}[\tau \cdot \cos(\mathbf{V}, \mathbf{T})]\} \tag{12}$$

where $\cos(\cdot,\cdot)$ represents the cosine similarity between two features.

D. Loss Function

1) Graph Loss: Let $\tilde{y}$ stand for ground-truth labels, a graph loss function is used to reward correct classifications and penalize misclassifications based on the sampled edges, as follows:

$$\mathcal{L}_{\text{graph}} = \sum_{\substack{m=1\ldots s \\ n:(m,n)\in\varepsilon}} \delta\left(y_m, \tilde{y}_m\right) \log \mathbf{P}_{m,n} \tag{13}$$

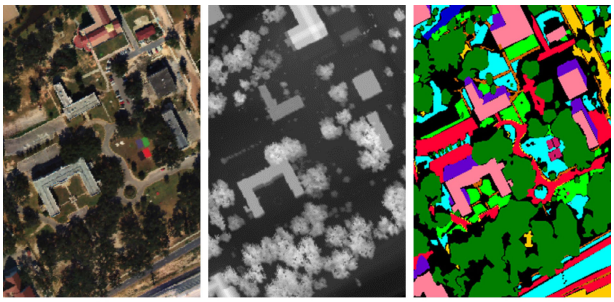
where $\delta(\cdot,\cdot)$ is a reward function that quantifies the discrepancy between the average accuracy (AA) of the mth sample and the current success value which is 1 if $y_m = \tilde{y}_m$ and 0 otherwise.

2) Contrastive Alignment Loss: The total contrastive loss for aligment of text–visual features is defined as

$$\mathcal{L}_{\text{CA}} = \frac{(\mathcal{L}_{T\to V} + \mathcal{L}_{V\to T})}{2} \tag{14}$$

which is composed of a text-to-vison contrastive loss defined as

$$\mathcal{L}_{T\to V} = -\frac{1}{c}\sum_{m=1}^{l}\sum_{n=1}^{c} \log\left(\frac{\exp\left(\tau \cdot \cos\left(\mathbf{T}_n^m, \mathbf{V}_n^m\right)\right)}{\sum_{k=1}^{c}\exp\left(\tau \cdot \cos\left(\mathbf{T}_n^m, \mathbf{V}_k^m\right)\right)}\right) \tag{15}$$



(a) (b) (c)

Fig. 5. MUUFL dataset. (a) Pseudocolor image for HSIs. (b) LiDAR DSM map. (c) Groundtruth map.

TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE MUUFL DATASET

Class		Nunber of samples	
No	Name	Train	Test
1	Trees	150	23246
2	Mostly grass	150	4270
3	Mixed ground	150	6882
4	Dirt and sand	150	1826
5	Roads	150	6687
6	Water	150	466
7	Buildings shadows	150	2233
8	Buildings	150	6240
9	Sidewalks	150	1385
10	Yellow curbs	150	183
11	Cloth panels	150	269
total		1650	53687

and a vison-to-text contrastive loss defined as

$$\mathcal{L}_{V\to T} = -\frac{1}{c}\sum_{m=1}^{l}\sum_{n=1}^{c} \log\left(\frac{\exp\left(\tau \cdot \cos\left(\mathbf{V}_n^m, \mathbf{T}_n^m\right)\right)}{\sum_{k=1}^{c}\exp\left(\tau \cdot \cos\left(\mathbf{V}_n^m, \mathbf{T}_k^m\right)\right)}\right). \tag{16}$$

By integrating all the mentioned losses and the fusion-contrast loss from (4), the final objective function for the end-to-end training of TMCFN is given by

$$\mathcal{L} = \mathcal{L}_{FC} + \mathcal{L}_{\text{CA}} + \mathcal{L}_{\text{graph}}. \tag{17}$$

IV. EXPERIMENTS

A. Experiment Setup

1) Dataset Description: We evaluate the performance of various methods on three publicly available datasets consisting of HSIs and corresponding LiDAR-derived digital surface model (DSM) data: MUUFL [46], Houston [47], and Trento [17].

The MUUFL dataset was collected on the campus of the University of Southern Mississippi, obtaining HSIs and LiDAR data. HSIs consist of 64 available wavelength bands ranging from 375 to 1050 nm. Both HSIs and LiDAR data have 325 × 220 pixels, with a spatial resolution of 1 m. This dataset contain 11 classes and a total of 53 687 samples. Fig. 5 shows the visualization of the MUUFL dataset. The numbers of training and testing samples are shown in Table I.

The Houston dataset was acquired in 2012, covering the University of Houston campus and surrounding urban areas.

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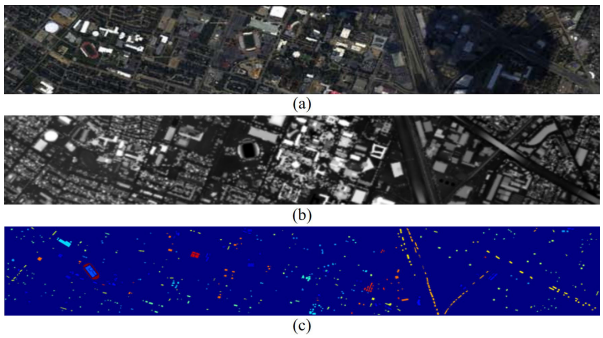


Fig. 6. Houston dataset. (a) Pseudocolor image for HSIs. (b) LiDAR DSM map. (c) Groundtruth map.

TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE HOUSTON DATASET

Class		Number of samples	
No	Name	Train	Test
1	Health grass	180	1251
2	Stressed grass	180	1254
3	Synthetic grass	180	697
4	Trees	180	1244
5	Soil	180	1242
6	Water	180	325
7	Residential	180	1268
8	Commercial	180	1244
9	Road	180	1252
10	Highway	180	1227
11	Railway	180	1235
12	Parking lot 1	180	1233
13	Parking lot 2	180	469
14	Tennis court	180	428
15	Running track	180	660
total		2700	15029

Both HSIs and LiDAR data consist of 349 × 1905 pixels, with a spatial resolution of 2.5 m. HSIs encompass 144 bands distributed in the 380–1050 nm region. This dataset comprises 15 object categories, totaling 15 029 samples. Fig. 6 shows the visualization of the Houston dataset. The numbers of training and testing samples are shown in Table II.

The Trento dataset was captured in a rural area near the city of Trento, Italy. The spatial sizes of both HSIs and LiDAR data are 166 × 600, and their spatial resolutions are 1 m. HSIs contain 63 bands, distributed in the 420–990 nm region. This dataset has six object categories with a total of 30 214 samples. Fig. 7 shows the visualization of the Trento dataset. The numbers of training and testing samples are shown in Table III.

2) Implementation Details: We implement the TMCFN using the PyTorch framework on Ubuntu. The model takes HSIs, LiDAR data and each class text description as input. For the MUUFL, Houston, and Trento datasets, we set the number of data subsets M to 15, 6, and 10, respectively. We uniformly set the patch size p to 19, and train our model on two NVIDIA GeForce 3090 GPUs for 1000 epochs using Adam as the optimizer with the learning rate for VFR setting to 0.0005. We load the pretrained parameters of the text encoder in CLIP into the text encoder of this article and fix the parameters. Given the variability in the structure of

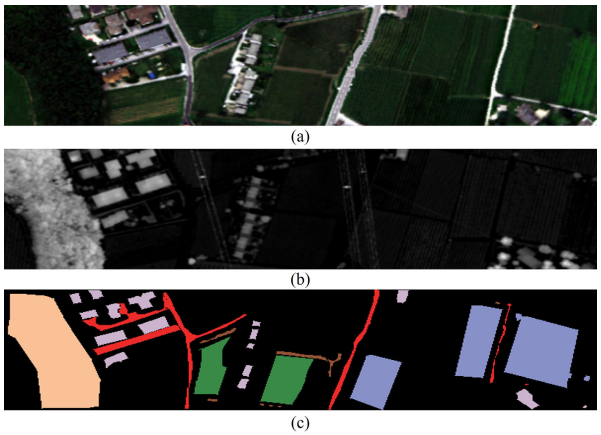


Fig. 7. Trento dataset. (a) Pseudocolor image for HSIs. (b) LiDAR DSM map. (c) Groundtruth map.

TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE TRENTO DATASET

Class		Number of samples	
No	Name	Train	Test
1	Trees	100	4034
2	Buildings	100	2903
3	Ground	100	479
4	Woods	100	9123
5	Vineyard	100	10501
6	Roads	100	3174
total		600	30214

graphs, we utilize GraphSAGE [48] as the GNNs to extract structural features due to its superior scalability. Note that we do not sample neighbors of a node. During the training and testing stage, all neighbors of a node will participate, and a learnable aggregation function will aggregate the information of all neighbors to update the features of the nodes. The code is available at https://github.com/Jiahuiqu/TMCFN.

B. Classification Accuracy and Discussion

To show that our method is advanced, we conduct comprehensive comparisons with eight state-of-the-art methods, including HCT [27], MFT [28], CCR-Net [49], AM^{3}Net [24], S^{2}ENet [50], EndNet [51], FusAtNet [52], and SVM [53]. Specifically, CCR-Net [49] obtains fused representations of multisource data by adopting a cross-modal reconstruction strategy. In S^{2}ENet [50], a spatial–spectral enhancement module is utilized to efficiently enhance the interaction of cross-modal information, improving classification performance of the model. EndNet [51] employs a deep encoder–decoder network for fusion of multimodal features and images classification. FusAtNet [52] leverages self-attention and cross-attention mechanisms to extract spatial–spectral features from HSIs and LiDAR data for classification. SVM [53] is an excellent representative of machine learning methods. HCT [27], MFT [28], and AM^{3}Net [24] were introduced earlier. The comparative methods are all based on discrete label supervision, providing a fair benchmark to assess the effectiveness of our proposed method. We evaluate the

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TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE MUUFL DATASET

Class	HCT	MFT	CCR-Net	AM^{3}Net	S^{2}ENet	EndNet	FusAtNet	SVM	Proposed
Trees	89.60	90.64	94.81	82.31	88.32	87.22	89.46	83.00	94.64
Mostly grass	87.28	79.16	75.11	61.55	88.10	80.91	82.62	81.45	94.26
Mixed ground	82.55	79.86	76.69	41.88	85.50	73.93	72.80	71.83	84.92
Dirt and sand	94.85	90.03	93.70	91.68	94.41	92.44	91.51	84.39	95.13
Roads	83.13	79.48	92.33	73.29	91.34	87.26	90.16	89.59	92.91
Water	99.79	99.57	100.00	100.00	100.00	99.14	100.00	99.14	100.00
Buildings shadows	94.89	94.54	92.07	94.09	94.58	91.04	96.15	88.94	96.33
Buildings	94.87	92.44	95.00	82.80	97.12	93.73	95.99	87.69	95.35
Sidewalks	78.56	77.11	69.68	61.23	86.57	81.73	82.74	73.72	87.22
Yellow curbs	98.36	96.72	91.26	94.54	98.91	99.45	96.72	99.45	100.00
Cloth panels	98.88	99.26	96.65	98.51	98.88	100.00	99.63	98.51	99.63
OA (%)	88.60	87.10	89.88	74.95	89.95	86.18	87.97	83.14	**93.22**
AA (%)	91.16	88.98	88.85	80.17	93.07	89.72	90.71	87.07	**94.58**
$\kappa \times 100$	85.29	83.32	86.73	68.06	87.07	82.22	84.52	78.51	**91.14**

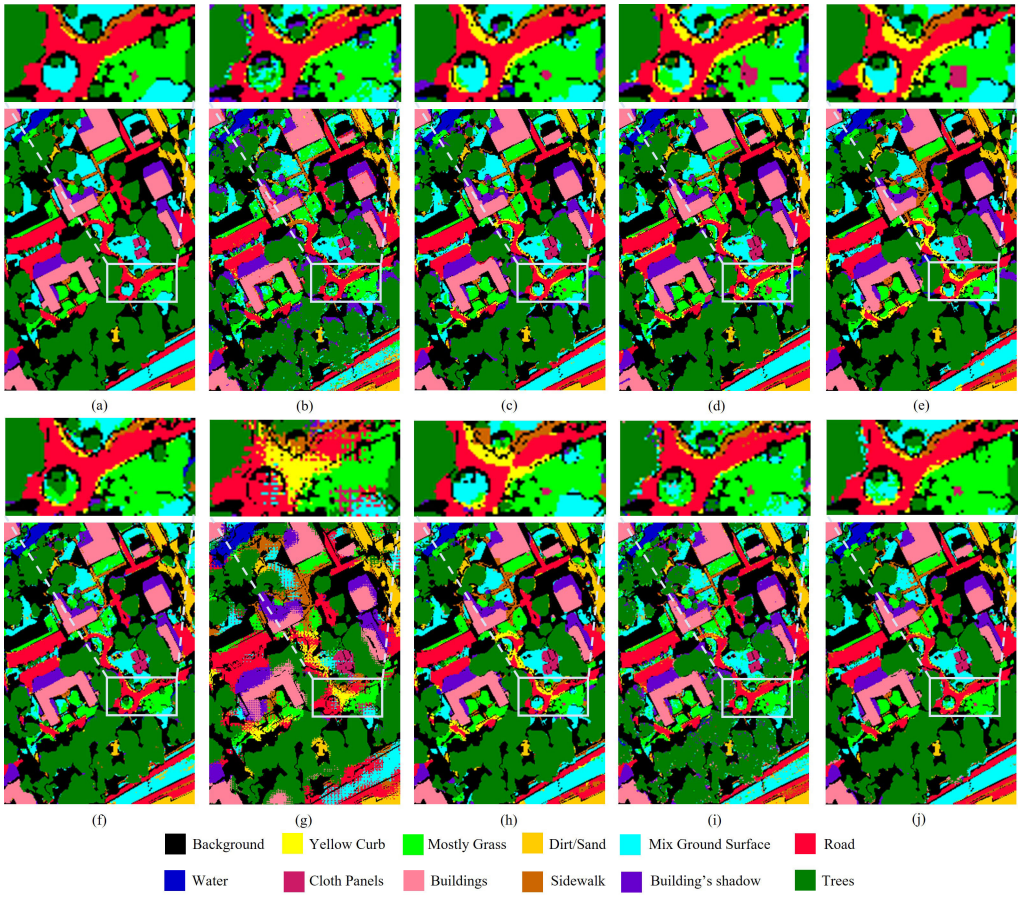


Fig. 8. Classification result maps for different comparison methods on the MUUFL dataset. (a) Groundtruth. (b) SVM. (c) s^{2}ENet. (d) FusAtNet. (e) MFT. (f) CCR-Net. (g) AM^{3}Net. (h) HCT. (i) EndNet. (j) Proposed.

performance of all methods through three metrics: overall accuracy (OA), AA, and Kappa Coefficient (κ).

Table IV shows the classification results of different methods on the MUUFL dataset. Our proposed TMCFN achieves the highest OA (93.22%), AA (94.58%), and κ (91.14) compared to other methods, attaining the highest accuracy in seven categories (mostly grass, dirt and sand, roads, water, buildings shadows, sidewalks, and yellow curbs), with mostly grass (94.26%) significantly outperforming other methods. Classification result maps for all methods on the MUUFL dataset are shown in Fig. 8. Combining with the overall images and local enlarged images, it can be seen that the proposed TMCFN achieves favorable results even in scenes with complex distribution of buildings, which shows the capability of TMCFN to effectively integrate the rich heterogeneous information available in HSIs and LiDAR data under the guidance of text information.

Table V shows the classification results of different methods on the Houston dataset. As can be seen from Table V, compared to other methods, the proposed TMCFN demonstrates

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TABLE V
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HOUSTON DATASET

Class	HCT	MFT	CCR-Net	AM^{3}Net	S^{2}ENet	EndNet	FusAtNet	SVM	Proposed
Health grass	99.52	99.52	97.60	94.00	98.72	99.60	92.57	95.68	99.52
Stressed grass	99.76	98.80	99.68	95.69	99.76	98.33	97.29	98.56	100.00
Synthetic grass	99.71	99.71	100.00	98.85	100.00	100.00	99.71	99.57	100.00
Trees	99.68	99.84	99.76	94.61	99.12	97.75	98.79	98.71	99.92
Soil	100.00	100.00	99.92	99.36	100.00	98.55	97.99	98.87	100.00
Water	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.38	100.00
Residential	99.21	99.21	98.74	95.43	98.19	97.08	97.24	86.28	98.90
Commercial	97.83	99.28	93.65	97.27	98.23	99.12	94.37	90.76	99.60
Road	97.92	97.84	83.31	92.25	97.04	93.13	92.33	79.63	99.28
Highway	100.00	99.67	99.19	99.02	98.53	98.21	95.11	87.45	100.00
Railway	100.00	100.00	97.57	99.27	99.43	95.71	98.14	88.83	99.60
Parking lot 1	99.68	99.76	99.11	92.62	97.65	98.38	94.24	77.13	99.59
Parking lot 2	100.00	98.93	85.50	99.79	97.44	92.96	95.74	67.16	99.79
Tennis court	100.00	100.00	100.00	100.00	100.00	99.30	99.07	100.00	100.00
Running track	100.00	100.00	100.00	98.94	100.00	100.00	99.85	99.85	100.00
OA (%)	99.45	99.45	96.93	96.53	98.82	97.76	96.35	90.82	**99.69**
AA (%)	99.55	99.50	96.93	97.14	98.94	97.87	96.83	91.19	**99.75**
$\kappa \times 100$	99.41	99.40	96.68	96.25	98.73	97.58	96.41	90.07	**99.67**

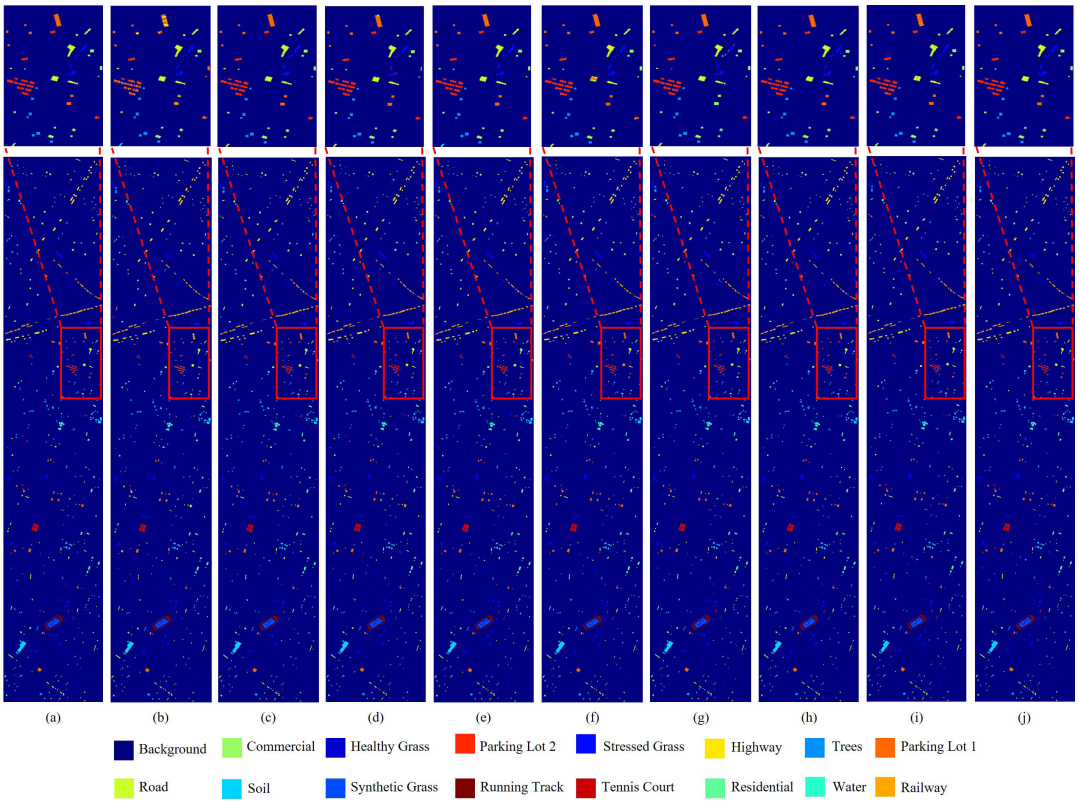


Fig. 9. Classification result maps for different comparison methods on the Houston dataset. (a) Groundtruth. (b) SVM. (c) S^{2}Enet. (d) FusAtNet. (e) MFT. (f) CCR-Net. (g) AM^{3}Net. (h) HCT. (i) EndNet. (j) Proposed.

the top performance in terms of OA (99.69%), which is 0.24% higher than both HCT and MFT, and 0.87% higher than S^{2}ENet. In addition, TMCFN exhibits superior classification accuracy in ten categories. In the complex and diverse Houston dataset, TMCFN demonstrates excellent classification performance, making it highly adaptable for application in various complex scenarios. Classification result maps for all methods on the Houston dataset are shown in Fig. 9. As can be seen in Fig. 9, combining the overall image and the local enlarged image, the proposed TMCFN achieves the best classification performance, with a more comprehensive presentation and a lower noise level for each category, followed by HCT and MFT.

Table VI shows the classification results of all methods on the Trento dataset, our proposed TMCFN achieves the best OA (99.36%), AA (99.32%), and κ (99.15%), particularly excelling in the classification accuracy of the Trees (99.75%), Buildings (99.14%), and Ground (99.58%). From Table III, we can observe that the number of training samples for each class in the Trento dataset is relatively low. Especially, the training samples for the Vineyard category constitute less than 1% of the total quantity. However, it can be seen

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TABLE VI
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE TRENTO DATASET

Class	HCT	MFT	CCR-Net	AM^{3}Net	S^{2}ENet	EndNet	FusAtNet	SVM	Proposed
Trees	99.53	98.49	98.91	97.12	99.23	89.37	98.34	92.46	99.75
Buildings	97.59	98.66	97.76	96.73	98.07	98.21	97.55	95.97	99.14
Ground	97.91	98.54	94.57	98.96	98.54	96.87	96.24	98.33	99.58
Woods	100.00	99.41	100.00	99.12	100.00	99.20	99.43	99.89	99.72
Vineyard	99.71	98.93	99.58	98.22	100.00	86.76	100.00	86.70	99.22
Roads	97.07	98.39	98.96	93.76	95.84	92.47	99.46	91.68	98.49
OA (%)	99.27	98.93	99.30	97.75	99.25	92.73	99.26	93.05	**99.36**
AA (%)	98.64	98.74	98.30	97.32	98.61	93.81	98.51	94.17	**99.32**
$\kappa \times 100$	99.02	98.57	99.06	97.00	99.00	90.38	99.01	90.82	**99.15**

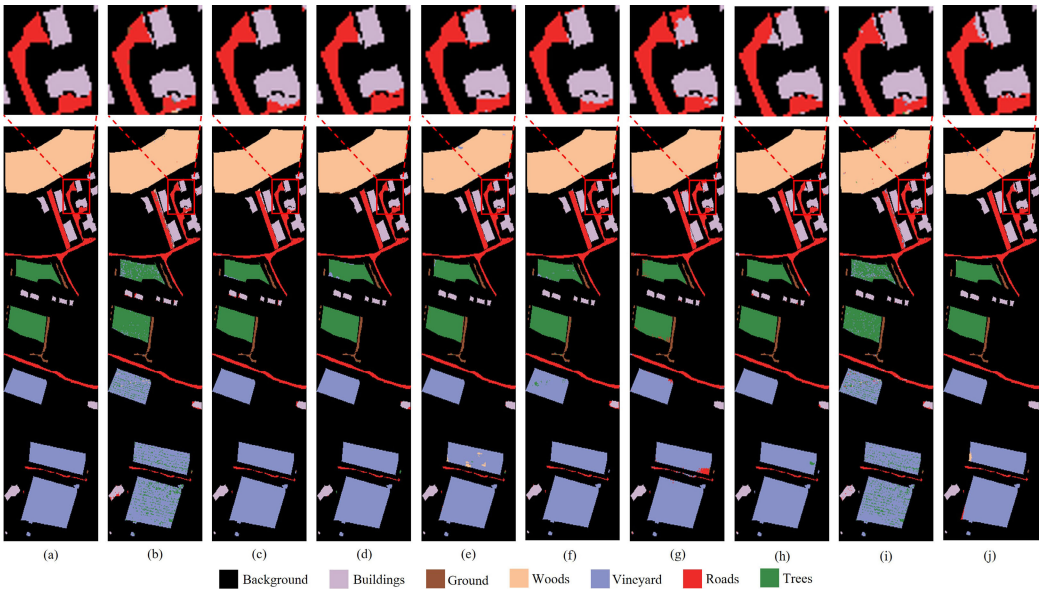


Fig. 10. Classification result maps for different comparison methods on the Trento dataset. (a) Groundtruth. (b) SVM. (c) S^{2}ENet. (d) FusAtNet. (e) MFT. (f) CCR-Net. (g) AM^{3}Net. (h) HCT. (i) EndNet. (j) Proposed.

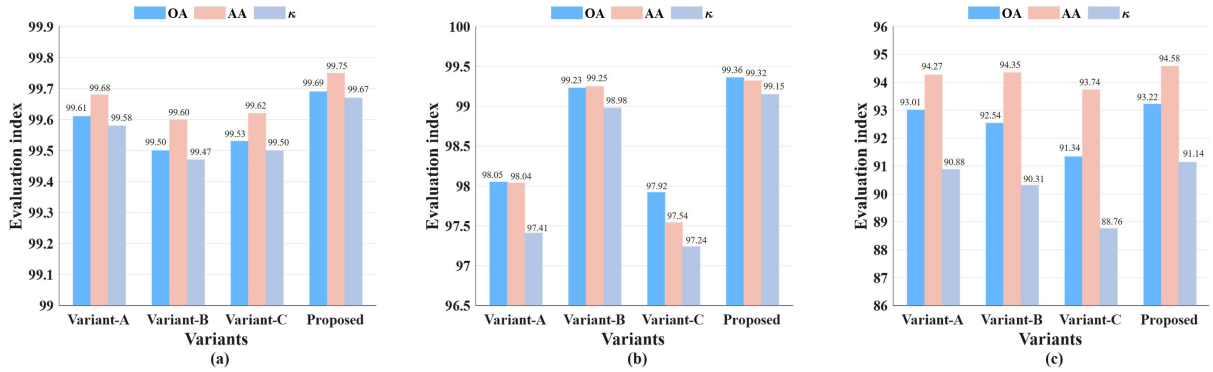


Fig. 11. Classification results of different variants for studying effectiveness of different component on (a) Houston dataset, (b) Trento dataset, and (c) MUUFL dataset.

from Table VI that TMCFN achieves satisfactory results in each category, which is mainly due to the carefully crafted FCFLM in the VFR of TMCFN using the self-supervised CL paradigm to alleviate the dependence on labeled samples. Classification result maps for all methods on the Trento dataset are shown in Fig. 10, it can be seen from the overall images that the proposed TMCFN correctly classifies the ground objects distributed in blocky space. Upon observing the local enlarged images, it is clear that the results of TMCFN exhibit less noise and roughness compared to the results of other methods.

C. Ablation Study

We carry out various ablation experiments to study the impact of different configurations on the performance of the proposed TMCFN.

1) Effectiveness of Different Components: To verify effectiveness of each component, four variants of TMCFN are experimented on the Houston, Trento, and MUUFL datasets. Specifically, Variant-A is generated by excluding DGSGM of dynamic graph-based feature learning. Variant-B is produced by replacing TAF^{2}M with a operation of concatenation. Variant-C is obtained by substituting FCFLM with

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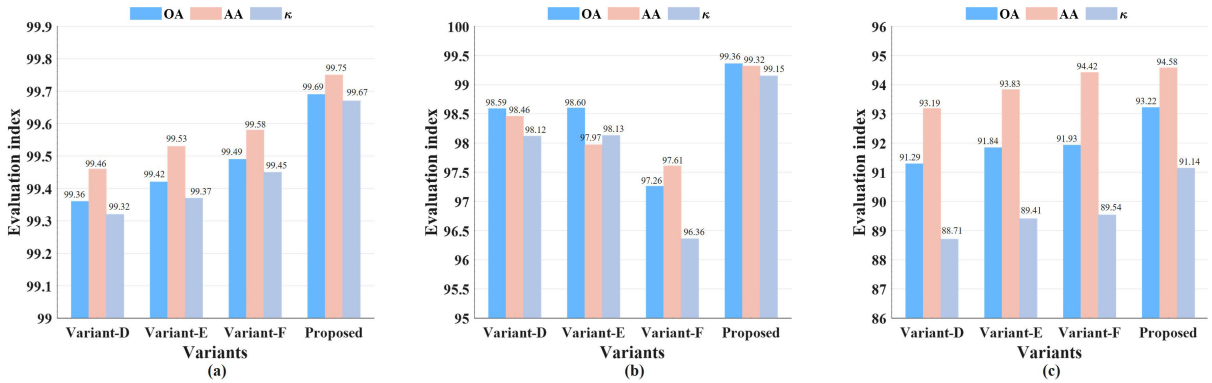


Fig. 12. Classification results of different variants for studying effectiveness of the FCFLM structure on (a) Houston dataset, (b) Trento dataset, and (c) MUUFL dataset.

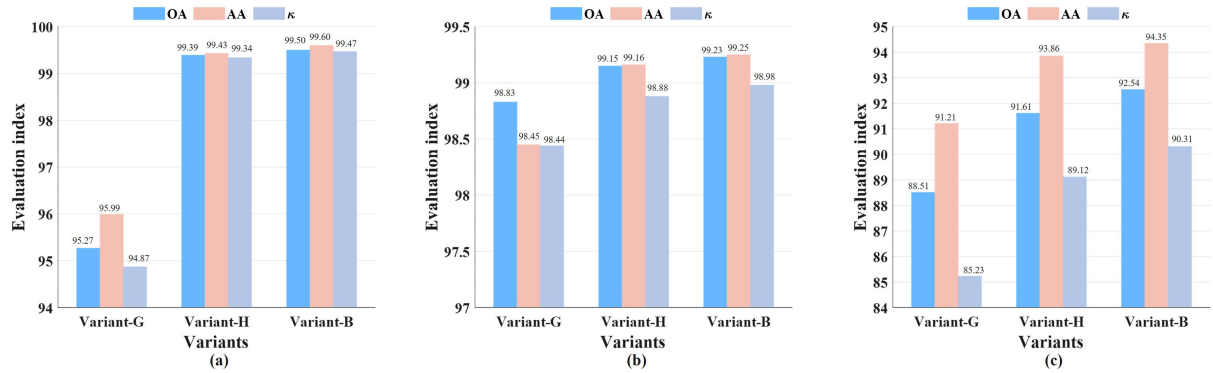


Fig. 13. Classification results of different variants for studying effectiveness of supervision forms on (a) Houston dataset, (b) Trento dataset, and (c) MUUFL dataset.

a shared ViT. Fig. 11 shows the classification accuracy of different variants on the Houston, Trento, and MUUFL datasets. We note that the performance of TMCFN surpasses that of each variants. To be specific, TMCFN outperforms Variant-A on the Houston, Trento, and MUUFL datasets in terms of OA, AA, and κ, which indicates that GNNs can extract more comprehensive topological features for classification from dynamic graph structures compared to a fixed graph structure. Moreover, TMCFN is superior to Variant-B by 0.19%, 0.13%, and 0.68% higher in terms of OA on the Houston, Trento, and MUUFL datasets, respectively, proving that TAF^{2}M can effectively integrate heterogeneous features from multimodal images under the guidance of text information. Besides, TMCFN performs superiorly to Variant-C, with improvements of 0.16%, 1.44%, and 1.88% in terms of OA on the Houston, Trento, and MUUFL datasets, respectively, showing that FCFLM can effectively promote latent multimodal representation learning.

2) Effectiveness of FCFLM Structure: To further explore the effectiveness of FCFLM structure in the proposed TMCFN, three variants of TMCFN are experimented on Houston, Trento, and MUUFL datasets. Specifically, Variant-D is obtained by replacing FCFLM with BYOL based on TMCFN. Variant-E is derived by replacing AFFM of FCFLM with the operation of concatenation. Variant-F is created by modifying the form of data input in FCFLM that two of the $\{\mathbf{H}_v^1, \mathbf{H}_v^2, \mathbf{H}_v^3\}$ are input into O-Net and the remaining one sample is input into T-Net. Fig. 12 depicts the classification performance of different variants. From the experimental results in Fig. 12, we can learn that TMCFN is 0.33%, 0.77%, and 1.93% higher than Variant-D in terms of OA on the Houston, Trento, and MUUFL datasets, respectively, demonstrating that FCFLM is superior to BYOL in extracting fused information between multimodal images. Moreover, the classification results obtained by the proposed method are better than those of Variant-E, providing evidence of the effectiveness of AFFM. Last but not least, FCFLM of the proposed method regards fused features of multimodal images from the T-Net as the regression target for training O-Net, while the corresponding structure of Variant-F uses unimodal features as the regression target, and experimental results show that the former is more conducive to multimodal representation learning. The aforementioned experiments substantiate the effectiveness of internal structure of FCFLM.

3) Effectiveness of Text Supervision: To further study effectiveness of text supervision, we utilize Variant-B (without TAF^{2}M) as the text-supervised baseline model, and also design two additional variants based on Variant-B, namely Variant-G and Variant-H. Specially, Variant-G is obtained by modifying Variant-B, replacing the supervision form from text supervision to discrete labels supervision. Variant-H uses non-extended text (class names) to guide training of VFR. We conduct experiments on the Houston, Trento, and MUUFL datasets using these variants. The results are shown in Fig. 13. It is evident that the text-supervised model

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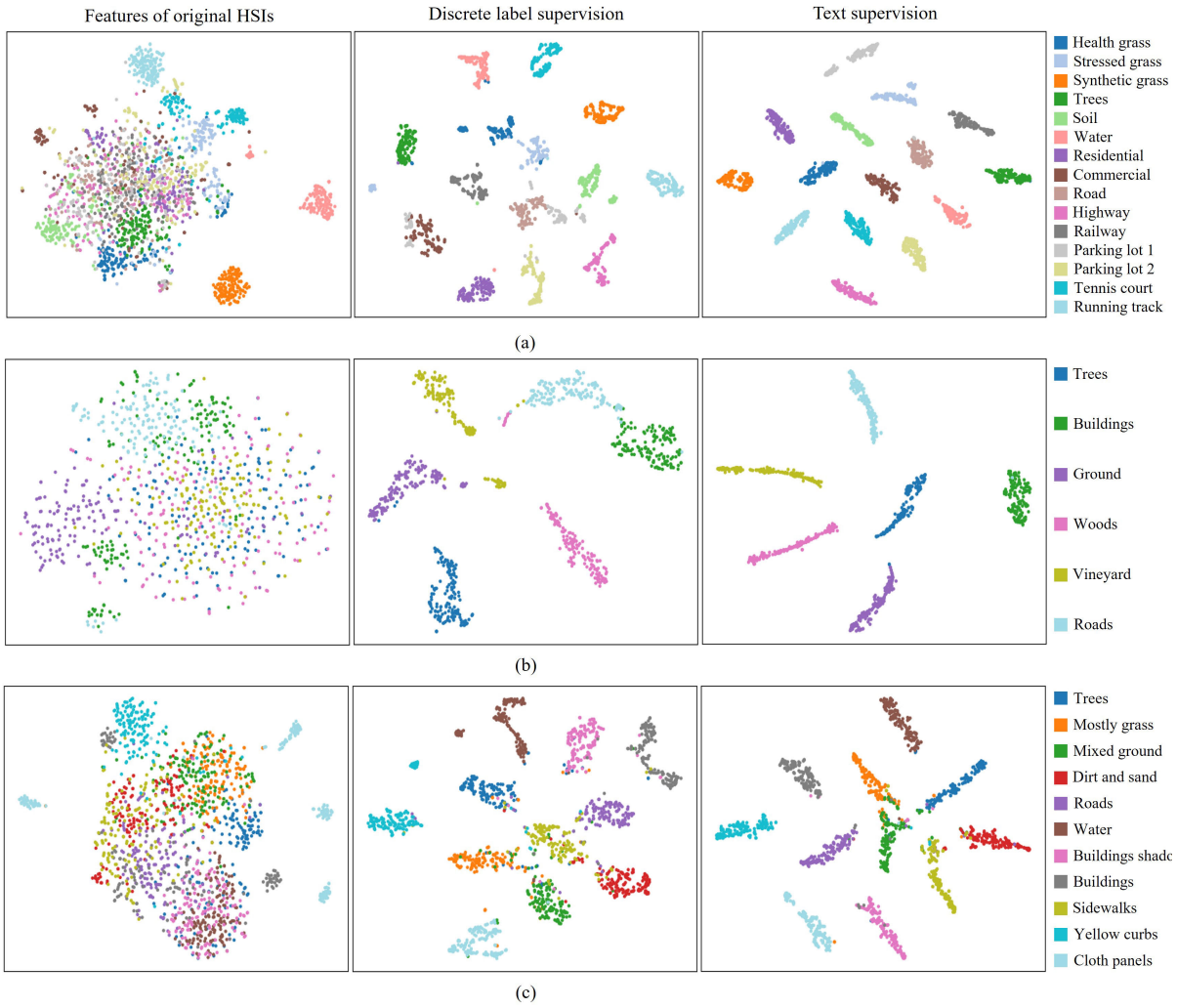


Fig. 14. Visualization of feature maps obtained from models with different supervision forms. The left image shows the feature map of the HSIs after PCA, the middle image shows the feature map from the model with discrete label supervision (Variant-G), and the right image shows the feature map from the model with text supervision (Variant-B). (a) Results on Houston dataset. (b) Results on Trento dataset. (c) Results on MUUFL dataset.

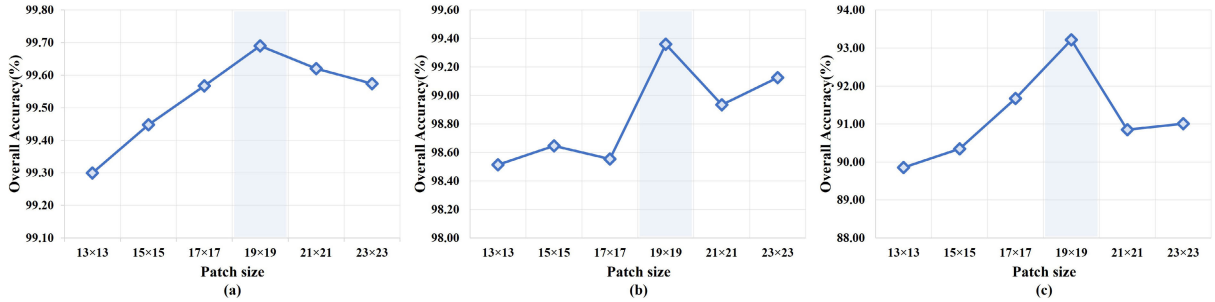


Fig. 15. Effect of image patch size on classification performance based on our proposed method using (a) Houston dataset, (b) Trento dataset, and (c) MUUFL dataset.

exhibits superior classification performance compared to the model supervised with discrete labels. Variant-H outperforms Variant-G by 4.10%, 0.32%, and 3.10% in terms of OA on the Houston, Trento, and MUUFL datasets, respectively, and Variant-B surpasses Variant-G by 4.23%, 0.4%, and 4.03% in terms of OA on the Houston, Trento, and MUUFL datasets, respectively. The results also show that the extended text is advantageous for the training of the network. In addition, we visualize the classification features of Variant-B and Variant-G, as shown in Fig. 14. We find that the visualization results of features obtained from Variant-B are better than those obtained from Variant-G. The features with the same class are more concentrated and the features with different classes are more discriminative in the former visualization

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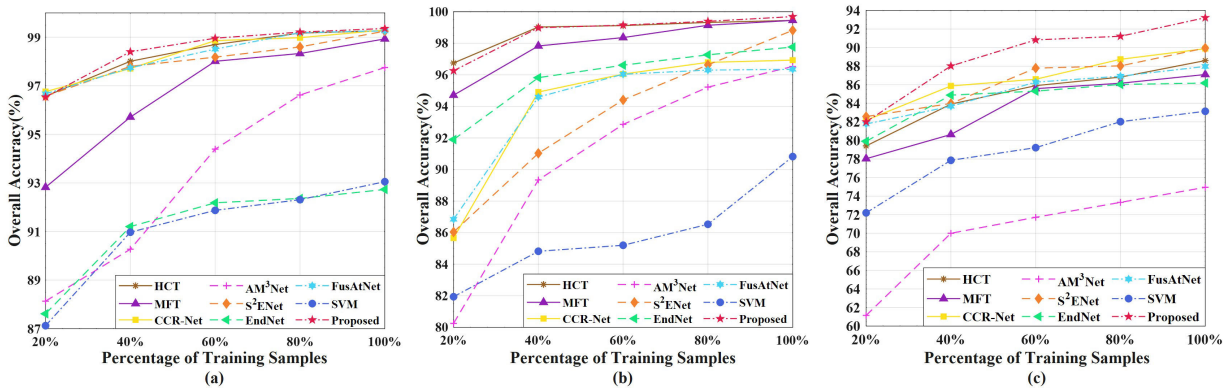


Fig. 16. Classification performance of our proposed TMCFN with varying percentages of training samples on (a) Houston dataset, (b) Trento dataset, and (c) MUUFL dataset.

results. This is mainly attributed to that TMCFN establishes text–visual association through CL, in which corresponding pairs of text–visual features are aligned and noncorresponding pairs are pushed away.

4) Effect of Image Patch Size: In this article, we construct n×n image patches centered on each pixel as the input for our proposed method. The selection of image patch size is important for classification tasks. Accordingly, we study the effect of different patch sizes on the performance of our method. Fig. 15(a)–(c) show the classification results obtained with various patch sizes (13×13, 15×15, 17×17, 19×19, 21×21, and 23×23) on the Houston, Trento, and MUUFL datasets, respectively. From Fig. 15, as the patch size increases from 13 to 19, the model performance tends to improve, which is primarily because larger patches can encompass more features at different scales and directions, enabling the model to better capture complex image features. However, the patch size is not always positively correlated with the model performance, which begins to decline when the patch size exceeds 19. This is mainly because an excessively large patch may introduce pixels with categories different from the pixel at the center of the patch, acting as noise that affects the feature representation of the central pixel. Based on the above analysis, the optimal patch size is 19×19.

5) Effect of Training Sample Numbers: In this section, we set up experiments to study the classification performance of various methods across different numbers of training samples to analyze the robustness of the proposed TMCFN to the number of training samples. The experimental results are depicted in Fig. 16. The horizontal axis in Fig. 16 represents the percentage of training samples (as listed in Tables I–III) changing from 20% to 100%, the vertical axis corresponds to the OA as an evaluation index. It can been seen from Fig. 16 that classification performance of all methods tends to improve as the percentage of training samples increases. Obviously, when the percentage of training samples is consistent, our proposed TMCFN generally outperforms other methods, demonstrating high-performance robustness to variations in the number of training samples. This can be principally attributed to meticulous extraction and fusion of latent multimodal features from HSIs and LiDAR data, alongside the establishment of a robust text–visual association.

TABLE VII

COMPARISON OF MODEL PARAMETERS AND EXECUTION TIME OF DIFFERENT METHODS ON THE TRENTO DATASET

Methods	MFT	AM^{3}Net	EndNet	TMCFN
Trainable Parameters(M)	0.32	2.80	0.08	0.72
Execution Time(s)	0.57	3.18	0.25	4.64

6) Efficiency Analysis of Different Methods: In this section, we train all methods under identical experimental conditions until they completely converge and reach optimality, and the number of parameters and time complexity are recorded. Table VII summarizes the number of trainable parameters and execution time of different methods. It can be seen from Table VII that the number of trainable parameters in the proposed TMCFN is at an intermediate level. Despite having the highest number of trainable parameters, AM^{3}Net does not outperform other methods in terms of classification performance. The probable reason lies in an excessive number of parameters leading to overfitting of the training data, thereby hindering effective generalization to test data. The proposed TMCFN achieves the best classification performance in comparative experiments with different methods, achieving a balance between classification performance and model complexity. The time complexity serves as another pivotal metric for describing the methods. We consider the time of inference for all test samples as execution time. As shown in Table VII, EndNet has the highest execution efficiency, but its classification accuracy is not better than all other methods. The execution time of our proposed TMCFN is less than that of MFT and EndNet, and basically equivalent to that of AM^{3}Net. The relatively long execution time of TMCFN primarily attributed to that the self-attention operation involved in transformer and the message passing mechanism involved in GNNs both have high-computational complexity. The classification performance of TMCFN is considerable and exhibits strong robustness, and the time complexity is worthwhile and acceptable.

V. CONCLUSION

In this article, we propose a TMCFN for joint classification of HSIs and LiDAR data, which capitalizes on the enriched semantic information provided by extended text to

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supervise the learning of visual representations and establish semantic connections with visual features. The classification is achieved by computing the similarity between visual features and each class text feature without an additional classifier. TMCFN yields superior classification results on three benchmark datasets, confirming the efficacy of the proposed network. The ablation experiments confirm the effectiveness of text supervision and the contributions of FCFLM and TAF^{2}M.

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